

31 Aug → Networking

1 Sep → Software

2 Sep → Microprocessor

3 Sep → Cyber Security

4 Sep → Physics
+ (11-12) Am

BUS

University of Asia Pacific

Admit Card
Examination of Spring, 2025

Financial Clearance

PAID



Department of Computer Science and

	COURSE TITLE	CR.HR.	EXAM. SCHEDULE
	Communication and Computer Networks	3.00	
	Communication and Computer Networks Lab	1.50	
	Software Engineering	3.00	
	Software Engineering lab	1.50	
4	CSE 315	Microprocessors and Microcontrollers	3.00
5	CSE 316	Microprocessors and Microcontrollers Lab	1.50
6	CSE 317	Computer and Cyber Security	3.00
7	BUS 301	Business and Entrepreneurship	3.00
8	PHY 301	Modern and Quantum Physics	3.00

Total Credit: 22.50

1. Examinees are not allowed to enter the examination hall after 30 minutes of commencement of examination for mid semester examinations and 60 minutes for semester final examinations.

2. No examinees shall be allowed to submit their answer scripts before 50% of the allocated time of examination has elapsed.

3. No examinees would be allowed to go to washroom within the first 60 minutes of final examinations.

4. No student will be allowed to carry any books, bags, extra paper or cellular phone or objectionable items/incriminating paper in the examination hall.
Violators will be subjects to disciplinary action.

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University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 2025
Program: B.Sc. in CSE
3rd Year 2nd Semester

Course Title: Data Communication and Computer Networks
Time: 1 hour

Credit Hour: 3.00

Course Code: CSE 311
Full Marks: 20

There are **Three** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Show the key differences between LAN, MAN, and WAN considering the following: size (coverage), data transfer speed, application, and typical use cases. [3]
- b. Briefly describe the seven layers of the OSI model, including functions and at least two protocols of each layer. [4]
- c. Discuss the differences between Layer 2 and Layer 3 devices. Define each device and their role in communication network from **Figure 1** given below: [3]

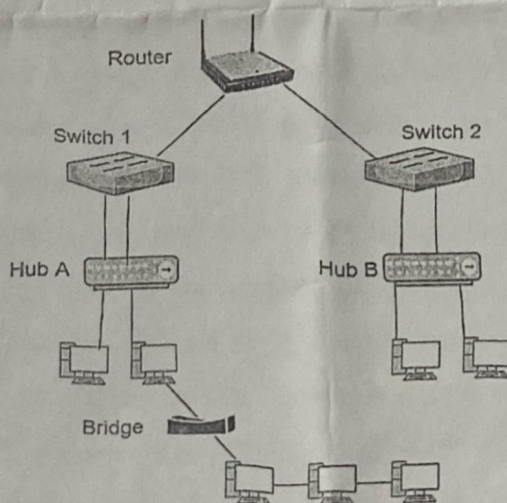


Figure 1. Basic Network devices by Layer

QUESTION 2 [10 MARKS]

- a. Why is line coding needed in digital communication? We have given a digital data bit as 010110101, you have to encode this data into the following line code scheme. [5]
- 1) Polar NRZ- L (Polar)
 - 2) Manchester (Dr. Thomas) (Polar)
 - 3) Alternate Mark inversion (AMI) (Bipolar)
- b. What is a subnet mask? Given IP Address: 172.16.50.200/19, find out the following: [5]
- a) Subnet mask.
 - b) Number of subnets.
 - c) Block size/host per subnet.
 - d) Number of valid hosts per subnet.
 - e) First and last valid host of each subnet.
 - f) Broadcast address.
 - g) Network address.

11100000
00110010
00100000

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Software Engineering
Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 313
Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. **Compare** static, dynamic, and decentralized webs in the context of web-based software engineering. [5]
- b. **Explain** the risk management process of software engineering [5]

QUESTION 2 [10 MARKS]

A national space agency is developing the onboard software system for a deep-space spacecraft mission. The system will control critical functions such as trajectory corrections, communication with Earth, fuel management, and onboard diagnostics. Since the spacecraft will operate millions of kilometers away, any defect could result in the complete failure of the mission, which has taken years of planning and billions of dollars of investment.

Before development begins, scientists and engineers have already specified the requirements in great detail. These requirements must be strictly followed, as they are based on well-studied physical models, safety regulations, and international standards. Each stage of the project must be thoroughly documented and approved before moving forward. Due to the critical nature of the mission, the software will only be tested when the entire system is fully developed, and partial or trial versions would not be useful.

As the project manager:

- a. Propose the most suitable software process model for this project from the following options: ~~RAD~~, ~~Waterfall~~, ~~Incremental~~, ~~Agile~~, V-model. **Examine** your answer and list the steps of that process. [5]
- b. **Generate** five most important functional requirements of the spacecraft software system based on the scenario above. [5]

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Fall 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Computer and Cyber Security
Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 317
Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Explain the ten security principles of cyber security. [5]
- b. Apply shift cipher to encrypt the given message where, [5]
 $K = (\text{last 2 digits of your student ID} \% 10) + 10$
Message = "UAP IS COMMITTED TO EXCELLENCE"

QUESTION 2 [10 MARKS]

- a. Describe the steps of the AES algorithm using the flowchart. [5]
- b. Sketch the Diffie-Hellman key exchange mechanism and find the shared secret key [5]
between Alice and Bob using the following data.

Alice

secret_key, $a=7$

public_key

$\{g=3, p=17\}$

Bob

secret_key, $b=11$

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 25
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Microprocessors and Microcontrollers
Time: 1 hour.

Credit Hour: 3.00

Course Code: CSE 315
Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. **Draw** the physical memory system of a microprocessor with a 24-bit data bus [4]
and 20-bit address bus, and calculate its total memory size.
- b. **1. Classify** the addressing mode of each instruction and **2. Calculate** the physical [2+4]
address where applicable (DS = 3000H).
- (i) MOV [4333H], AX
 - (ii) MOV [BX+4H], AX
 - (iii) MOV AL, 2564H
 - (iv) MOV CH, [BX]
 - (v) MOV AX, BX
 - (vi) MOV DX, [SI]

QUESTION 2 [10 MARKS]

You are given the following pseudocode:

Consider $a = AL$, $b = BL$, $c = CL$

Start

$a \leftarrow 3AH$

$b \leftarrow 7CH$

$c \leftarrow a - b$

$c \leftarrow c + 1$

$a \leftarrow \text{NOT } a$

$b \leftarrow -b$

swap(a, b)

$c \leftarrow c \oplus b$

$c \leftarrow c \vee b$

$c \leftarrow c \wedge a$

End

- a. **Modify** the given pseudocode into an equivalent Assembly program. [5]
- b. For each instruction in your assembly program, **determine** which flags (CF, ZF, SF, OF, PF, AF) are affected and show their status (0 or 1) after execution. [5]
Assume all flags are initially cleared (0) before execution.

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 2025
Program: B.Sc. in CSE
3rd Year 2nd Semester

Course Title: Modern and Quantum Physics

Time: 1 hour

Credit Hour: 3.00

Course Code: PHY 301

Full Marks: 20

There are **Two** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

a. State Planck's quantum hypothesis and explain blackbody radiation in terms of it. [5.0]

b. Show that the energy of an electron for an atom in n^{th} orbit [3.0]

$$E_n = -\frac{me^4Z^2}{8\epsilon_0^2n^2h^2}$$

where the symbols have their usual meanings.

c. Show that for hydrogen atom the total energy of an electron in ground state is [2.0]
-13.6 eV.

QUESTION 2 [10 MARKS]

a. Describe photoelectric effect. Mention the experimental observations of photoelectric effect. [3.0]

b. Explain Einstein's photoelectric effect equation. [2.0]

c. A photon of wavelength 400 nm strikes a metal surface with a work function of 2.2 eV. Calculate the kinetic energy of the emitted electron. (Planck's constant $h = 6.63 \times 10^{-34}$ Js; Speed of light $c = 3.0 \times 10^8$ m/s, $1 \text{ eV} = 1.6 \times 10^{-19}$ J) [5.0]

$$\frac{2eL}{9 \times 10^{-19} \times 400 \times 10^{-9}} = \frac{h}{\lambda} \times \frac{9 \times 10^{-19} \times 400 \times 10^{-9}}{\lambda \times h}$$
$$\frac{2eL}{E_0} = \frac{h}{\lambda \times h}$$
$$b = \frac{h \times 400 \times 10^{-9}}{2eL \times \lambda}$$

$$E = h\nu_0 + \frac{1}{2}mv_{\text{max}}^2$$

University of Asia Pacific
Department of Computer Science and Engineering
Mid Semester Examination, Spring 25
Program: B.Sc. in CSE
3rd year 1st Semester

Course Title: Business and Entrepreneurship
 Time: 1 hour.

Credit Hour: 3.00

Course Code: BUS 301
 Full Marks: 20

(Please return the question with the answer scripts)

Part A
(mandatory)

- 1 A company is considering three projects (A, B, and C) that are mutually inclusive.

Year	Project A BDT	Project B BDT	Project C BDT	
0 (Initial Investment)	(240,000)	(360,000)	(300,000)	300,000
1	100,000	120,000	140,000	440,000 140,000
2	100,000	130,000	150,000	520,000 220,000
3	100,000	140,000	160,000	750,000 160,000
4	100,000	160,000	180,000	930,000 180,000

Conditions & Constraints:

- The total budget for selected projects must not exceed BDT 600,000.
- The company wants to select projects based on the NPV while maintaining the Payback Period (if NPVs are similar).

$$PPC = 2 + \frac{159,000}{110,000} = 2.33$$

The discount rate is 11%.

- Calculate the **Payback Period** of all the projects. Select the project/s if the cutoff year is 4 3.5 years.
- Calculate the **net present value (NPV)** of all the projects. Select the project/s based on 4 NPV criteria.

Part B

Answer any of the three questions

- "A business idea starts with the proper analysis of needs, wants, and demands." Explain this statement with examples. 2
- "Company is an artificial person"- please describe 2
- Define Partnership business and explain its different types of partners. 2
- Differentiate between Industry and Commerce. 2

360,000

1	120,000	120,000
2	130,000	250,000
3	140,000	390,000
4	160,000	550,000

$$PPB = 2 + \frac{307,000}{190,000} = 2.22$$

240,000

1	100,000	100,000
2	100,000	200,000
3	100,000	300,000
4	100,000	400,000

$$PP = 2 + \frac{100,000}{300,000} = 2.2$$

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- e. As per the Companies Act, 1994, explain the steps for the formation of a private limited company. 2

Part C

Answer any of the three questions

- 3 a. Kamal deposited BDT 500,000 into a savings account that earns a monthly interest rate of 2% for 5 years. Calculate the future value (FV) after 5 years. 2
- b. You borrow BDT 200,000 at 10% annual interest, to be repaid over 5 years with equal annual payments. Prepare the loan amortization table. 2
- c. Mr. Rahman wants to take a loan of BDT 1,000,000. SCB Bank offers the loan at 10% annual interest, while HSBC offers it at 5% semi-annually. 2

Which loan option should Mr. Rahman choose if he wants to minimize his annual installment?

- d. John contributes \$5,000 annually for 20 years into an account earning 8% annual compound interest. After 20 years, he stops contributing but leaves the money invested for another 10 years. 2

What will be the total amount available for withdrawal at the end of 30 years.